

Andrei-Lucian Ștefan

Robotics Engineer (B.Eng., TUCN 2026) · Applied AI & Autonomous Systems

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PERSONAL WEBSITE: <https://lucianstefanandrei.github.io>

Graduate in **Robotics Engineering (B.Eng., TUCN, 2026)** with a multidisciplinary background in automation, programming, electronics, mechanics and system integration. I have hands-on experience in developing complex systems that combine mechanical, electrical and software components.

Specialized in Software Development, particularly **Python, C++, and Artificial Intelligence**, with a strong understanding of AI best practices. I have worked with **ROS 2, embedded systems, sensors, actuators, CAD, industrial platforms and embedded computing platforms**. Familiar with the complete engineering process, from design and prototyping to integration, testing, troubleshooting and technical documentation.

I am seeking a **junior role in engineering, software development, automation**, industrial systems, hardware-software integration, AI software development, manufacturing or technical maintenance, where I can apply a broad technical foundation and a practical hands-on approach.

FLAGSHIP PROJECT — BACHELOR'S THESIS

Development of an Autonomous Anthropomorphic Museum-Guide Robot

TUCN · Coordinator: Prof. Dr. Ing. Călin Neamțu · 2025–2026 · Demo & write-up on LinkedIn (see profile)

Designed and built a 1.3 m anthropomorphic robot for autonomous guidance in museums, featuring conversational artificial intelligence. The total component cost was multiple times lower than most of the commercial robots analyzed, at just **€4,357**.

Artificial Intelligence and Software

- **Multilingual conversational system** based on speech recognition, LLM processing, and speech synthesis, with **real-time response playback** to reduce perceived waiting time
 - **Comparison of local AI solutions and cloud-based services** in terms of cost, speed, privacy, and response quality
 - **Python/PySide6 graphical user interface** with multiple integrated functionalities, implemented as a ROS 2 node, together **with a ROS 2 Humble architecture deployed on NVIDIA Jetson boards**
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Navigation, Mechanics, and Power

- **Autonomous navigation** based on SLAM and Nav2, using a **Livox Mid-360 LiDAR, Orbbec Gemini 336 depth camera**, and ultrasonic sensors, with motor controlled by Teensy 4.1
 - **Modular 3D-printed structure, validated through FEA**, including the design and verification of the transmission system and shafts.
 - **LiFePO4 battery system** sized based on measured power consumption.
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SELECTED PROJECTS (AVAILABLE ON <https://lucianstefanandrei.github.io>)

- **Amnesia (June 2026 – Present)**. An open-source application designed as a secure, private, and offline journal using local AI assistants, RAG, application-level encryption, fast vector search and a desktop interface

- **Simulation of a robotic gear manufacturing process using RoboDK**
- **Mechanical Design of a worm gearbox**, including technical drawings, technical documentation and calculation
- **Feedback and validation for eStudent.ro (March 2025 – June 2025)** – Provided institutional-level feedback for the testing and validation of the modern eStudent digital learning and academic administration platform.

SKILLS

Programming	Python (primary) · C++ (strong algorithmic base and embedded systems) · LUA · SQL · MATLAB · Frontend (React, JavaScript, HTML, CSS), Node.js, Web Design ,
AI and ML	Data Analysis, Data Classification, local and cloud-based LLM/TTS integration, emotion recognition, Prompt Engineering
Robotics	ROS 2 Humble, Mechanical Design, Electrical Design, sensors, actuators, technical documentation
Embedded & Systems	NVIDIA Jetson (JetPack), ESP32, Teensy, Arduino, Linux, Git
CAD / Mechanical	Fusion 360, SolidWorks (Associate-certified), CATIA V5, AutoCAD, Autodesk Inventor, 3D Printing, G-Code, CNC
Automation	TIA Portal, Factory I/O, HMI, PLC, PID Control, RoboDK, DELMIA
Analytical & Soft	Structured problem decomposition, strong mathematical and algorithmic problem-solving capabilities, technical writing & documentation, self-directed learning across mechanical–electrical–software domains, bilingual communication (RO/EN, C2), teamwork.

EDUCATION

B.Eng. Robotics (EQF 6) — Technical University of Cluj-Napoca · 2022–2026 (graduated)

Faculty of Industrial Engineering, Robotics and Production Management. Coursework: Mechatronic Design, CAD, Flexible Manufacturing Systems, Automation of Production Lines, Embedded Programming, machine learning & computer vision, control & actuators, sensor integration, multi-platform programming.

CERTIFICATIONS, AWARDS (URL/ID ON LINKEDIN)

- **IBM Generative AI Engineering Professional Certificate** - IBM
- **Generative AI Leader Professional Certificate** - Google Cloud
- **TensorFlow Developer Certificate** - Google (2024, valid to 2027)
- **SolidWorks Associate** - Dassault Systèmes
- **Cybersecurity Analyst Professional Certificate** – IBM
- **Academia Inova 4.0 Contest – 2nd Place Robotics, 1st Place Energy Sustainability**

LANGUAGES

- **Cambridge C2 Proficiency** (English)
- **CILS B1 Italian Certificate** - Università per Stranieri di Siena
- **Languages:** Romanian (native) · English (C2, Cambridge) · Italian (B1, CILS) · French (B1)